

Communication and Data Transmission Mid Exam

Q1. Multiple Choice

1. In start-stop asynchronous method, the synchronization is found at:

a) the byte level

b) at the bits level

c) at stop level

d) at start level

2. Analog sinusoidal signal of 10V (p-p) has:

a) An infinite number of values

b) A definite number of values

c) Has only one value

d) Has only two values

3. the period of the electrical power in Jordan is equal to:

a) 20ms

b) 20 μ s

c) 17ms

d) $17\ \mu\text{s}$

4. What is the frequency of AC current which reaches zero value every 0.01 second?

a) 100 Hz

b) 50 Hz

c) 0

d) 500 Hz

5. A sinusoidal alternating current having a frequency of 50 Hz has a peak value of 10 A. What is the value of the current after $1/300$ s from zero?

a) 8.66 A

b) 181.5 mA

c) 50 A

d) 16.66 A

6. Two AC currents are given by: $i_1 = 50 \sin(314t + \pi/6)$ and $i_2 = 20 \sin(314t - \pi/4)$. Which of the two currents leads the other?

a) i_1 leads i_2 by 75°

b) i_2 leads i_1 by 75°

c) i_1 leads i_2 by 15°

d) i_2 leads i_1 by 15

7. Quantization noise in PCM, for a given input, is proportional to the:

- a) sampling rate
- b) maximum frequency of modulating signal
- c) the number of digits used for coding
- d) input signal

8. In a communication system, the noise is most likely to affect the signal:

- a) at the transmitter
- b) at the circuit
- c) at the channel
- d) at the destination

9. The human voice signal usually is:

- a) nonperiodic, composite with a limited range of frequencies
- b) periodic, uncomposite with a limited range of frequencies
- c) nonperiodic, composite with an unlimited range of frequencies
- d) periodic, uncomposite with an unlimited range of frequencies

10. Radio station in FM broadcasting has:

- a) 200 Hz bandwidth
- b) 200 kHz bandwidth

c) 200 MHz bandwidth

d) 200 GHz bandwidth

11. An amplifier has an output of 20w. What is the output in dBw:

a) 13 dBw

b) 130 dBw

c) 13dBmw

d) 130dBmw

12. If we want to transmit 1000 characters with each character encoded as 8 bits using asynchronous transmission, then the number of bits needed are:

a) 8000 bits

b) 10000 bits

c) 4000 bits

d) 5000 bits

13. Byte synchronization in synchronous transmission is accomplished by:

a) physical layer

b) data link layer

c) application layer

d) network layer

14. Forming bits into frames is performed by:

- a) physical layer
- b) data link layer
- c) application layer
- d) network layer

15. The interface circuit between your personal computer & wired telephone network is:

- a) Antenna
- b) Codec
- c) Modem
- d) Fiber

16. DSL stands for:

- a) Digital single line
- b) Digital system line
- c) Digital submission line
- d) Digital subscriber line

17. The most difficult medium to tape is:

- a) Wireless medium
- b) Fiber

c) Coax cable

d) Twisted pair

18. Given a channel with a capacity of 20 Mbps & Bandwidth of 3MHz. Assuming white noise, what SNR is required to achieve the capacity?

a) 101 dB

b) 102 dB

c) 100 dB

d) 103 dB

19. If an amplifier has 30 dB voltage gain, what voltage ratio does the gain represent?

a) 31.6 V

b) 31.6

c) 31.6 mV

d) 3.16

20. The power line voltage in Jordan can be expressed as:

a) $230 \sin(2 \cdot 501)$

b) $170 \sin(2\pi \cdot 60t)$

c) $230 \sin(2\pi \cdot 60t)$

d) $230 \sin(2\pi \cdot 360t)$

21. The most of energy in the signal is contained in:

- a) Spectrum
- b) Last harmonic
- c) Bandwidth
- d) Harmonics

22. Radio signals are made up of:

- a) Voltages and currents
- b) Electrons and protons
- c) Noise and data
- d) Electric and magnetic fields

23. The communications medium causes the signal to be:

- a) Interfered with
- b) Attenuated
- c) Amplified
- d) Modulated

24. If a signal has a DC component, then:

- a) It has $f=0$ & a non-zero average amplitude
- b) It has $f \neq 0$ & a non-zero average amplitude

c) It has $f \neq 0$ & a zero average amplitude

d) It has $f = 0$ & a zero average amplitude

25. Which of the following is correct:

a) The higher data rate of signal, the greater is the bandwidth & the greater the distortion

b) The higher data rate of signal, the greater is the bandwidth & the smaller the distortion

c) The higher data rate of signal, the smaller is the bandwidth & the greater the distortion

d) The smaller data rate of signal, the greater is the bandwidth & the greater the distortion

26. Delay distortion is caused by:

a) Different velocity of propagation for different harmonics

b) Constant velocity of propagation for different harmonics

c) Different velocity of propagation in free space for different harmonics

d) Constant velocity of propagation in free space for different harmonics

27. The following medium is limited in distance, bandwidth & data rate:

a) Coax cable

b) Waveguide

c) Twisted pair

d) Free space

28. The following mode is used in telephony:

- a) Step index multimode
- b) Graded index multimode
- c) Single index multimode
- d) Single mode propagation

29. Usually, human vision is allocated with respect to wavelength in:

- a) Megameters
- b) Nanometers
- c) Millimeters
- d) Micrometers

30. Radio waves travel more slowly near the ground, because:

- a) The refractive index of the atmosphere decreases with height
- b) The refractive index of the atmosphere increases with height
- c) The refractive index of the atmosphere is the same regardless of the height
- d) The reflection index of the atmosphere decreases with height

31. The combination of Tx/Rx in the satellite is called:

- a) Uplink
- b) Transponder

c) Downlink

d) Repeater

32. Multipath could be:

a) Could be an advantage and could be a disadvantage

b) Always an advantage

c) Always a disadvantage

d) Always a merit

33. International audio broadcasting is usually used:

a) Ground wave propagation

b) Direct wave propagation

c) Sky wave propagation

d) Surface wave propagation

34. Typical speech has a dynamic range of about:

a) 2.5 dB

b) 250 dB

c) 0.25 Db

d) 25 dB

35. Changing TV channels by using remote control is an example of:

- a) simplex point-to-point communication
- b) half-duplex point-to-point communication
- c) simplex multipoint communication
- d) half-duplex multipoint communication

36. Voice broadcasting mesh network contains 5 stations, where the traffic moves from 1 to 2, 2 to 3, 3 to 4 & so on. The number of links in this network are:

- a) 10
- b) 5
- c) 24
- d) 20

37. Network topology refers to the layout of:

- a) physically connected devices
- b) logically connected devices
- c) physically & logically connected devices
- d) physically & semi-logically connected devices

38. Multiple connections of devices in a network can be found in:

- a) mesh & star topologies

- b) bus only
- c) mesh & ring topologies
- d) bus & star topologies

39. End to end logic connection between application layers is done by:

- a) directly between application layers at both sides
- b) all layers
- c) directly between application & network layers at both sides
- d) logically between application & network layers at both sides

40. Power companies send a simple sine wave with a frequency of 60 or 50 Hz. This wave carries:

- a) power & energy
- b) energy
- c) power
- d) nothing

41. The actual measurement of the speed of the link is:

- a) bandwidth
- b) throughput
- c) data rate

d) range

42. The wavelength of 600THz green light ray in air is:

a) 0.5 μm

b) 0.05 μm

c) 0.005 μm

d) 500 μm

43. Radio signals are made up of:

a) voltages & currents

b) electric & magnetic fields

c) electrons & protons

d) noise & data

44. The capacity of a channel with 2MHz bandwidth & SNR = 3981 is:

a) 10.4 Mbps

b) 24 Mbps

c) 10.3 Mbps

d) 23.9 Mbps

45. The two criteria: throughput & delay are often contradictory, and this is due to:

a) traffic

b) traffic congestion

c) traffic delay

46. Baseband transmission is:

a) a single-channel system that supports single transmission

b) a multi-channel system that supports single transmission

c) a single-channel system that supports multi-transmission

47. A measure of the capacity of a circuit or channel:

a) SNR

b) bandwidth

c) bps

48. The value of the upper frequency of the voice channel in terms of wavelength is:

a) 75 km

b) 67.5 km

c) 75 m

49. The following network is a circuit-switched network:

a) Internet

b) satellite

c) telephony

50. Voice broadcasting mesh network contains 5 stations, the number of links in this network are:

a) 10

b) 5

c) 20

51. In a digital transmission, the sender clock is 0.2 percent faster than the receiver clock. How many extra bits per second does the sender send if the data rate is 1 Mbps?

a) 2000 bits

b) 200 bits

c) 20 bits

Q2. Problems

1. A power amplifier has an input of 90 mV across 10 k Ω . The output is 7.8 V across an 8 Ω speaker.

a) What is the power gain, in decibels?

b) If the power amplifier is connected to two other amplifiers with gains of 10 dB & 20 dB and two filters with attenuations of -2 dB & -4 dB, calculate the overall gain.

2. An amplifier rated at 72 w output is connected to an 8 Ω speaker.

- a) What input power is required for full power if the power gain is 30dB
 - b) What is the input voltage for the rated output if the amplifier voltage gain is 40dB?
3. Find the 8-bit data stream for each case depicted in Figure (The figure is on page 11 of the document).